

Digital Twins in Clinical Development: A Landscape Survey (2021-2026)

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Date: May 2026

Status: Draft — separating hype from reality

Executive Summary

Digital twins — computational models of individual patients used to simulate treatment responses — have become one of the most hyped concepts in clinical development. Vendors promise shorter trials, smaller control arms, and billions in savings. Regulators have responded with frameworks for AI credibility and model qualification. Meanwhile, most real applications remain in pilot settings, and the gap between the “full physiological twin” vision and what is actually feasible remains wide.

This survey maps the current landscape: regulatory guidance, vendor offerings, methodological approaches, case studies with real evidence, and an honest assessment of where the hype outruns the reality.

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1. What Is a Digital Twin in Clinical Development?

1.1 Defining the Terms

The terminology around digital twins in healthcare is muddled — different vendors and publications use the same words to mean different things. A clear taxonomy is essential before evaluating claims.

Term	Definition	Example
Digital Twin	A dynamic computational model of an individual patient that is updated with real-time data. Originates from NASA/engineering (physical asset + digital replica + data link).	None fully exist in clinical trials yet

Term	Definition	Example
Virtual Patient	A static computational model of a patient, typically built from historical data. Not dynamically updated.	Most of what vendors call “digital twins”
Synthetic Control Arm	A counterfactual cohort constructed from historical or real-world data to replace or supplement a randomized control group.	Phesi’s cGvHD twin, Unlearn’s placebo predictions
Prognostic Score	A single numerical summary of a patient’s predicted outcome under control, derived from a predictive model.	Any vendor-provided or internally developed score
External Control Arm (ECA)	Control group sourced from outside the trial (historical data, RWD), used when randomization is unethical or impractical.	FDA draft guidance on externally controlled trials
Model-Informed Drug Development (MIDD)	Umbrella term for using models in drug development; digital twins are a subset.	FDA MIDD Paired Meeting Program

Bottom line: What most vendors call “digital twins” in 2026 are actually virtual patient models or synthetic control arms — not true digital twins in the engineering sense. True digital twins (dynamic, data-linked, continuously updating) remain aspirational in healthcare.

1.2 The Value Proposition

The core idea is compelling: if you can accurately predict what would happen to a patient on control, you can:

1. **Reduce control arm size** — fewer patients exposed to placebo
2. **Increase power** — more precise treatment effect estimates
3. **Shorten trials** — faster enrollment, fewer dropouts
4. **Enable trials where randomization is unethical** — rare diseases, oncology, pediatrics
5. **Identify responders** — personalized treatment effect estimation

The question is: how well does this actually work in practice?

2. Regulatory Landscape

2.1 Key Milestones

Date	Agency	Action	Significance
Sep 2022	EMA	Qualified PROCOPA for continuous outcomes in Ph2/3 trials	First-ever regulatory qualification of a DT-derived methodology
Feb 2023	FDA	Draft guidance on externally controlled trials	Framework for when synthetic/EC arms may be acceptable

Date	Agency	Action	Significance
2023	EMA	PROCOVA qualification for phase 2/3 (official opinion)	Validated the concept commercially
Jan 2025	FDA	AI Credibility Framework (7-step risk-based process)	How to establish model credibility for regulatory use
2024 (draft)	ICH	M15 — global harmonized MIDD standards	Infrastructure for large-market expansion
Jan 2026	FDA/EMA	“Guiding Principles of Good AI Practice” (10 principles)	Joint alignment on AI in drug development
2025–26	FDA	Multiple ISTDND submissions (Unlearn, etc.)	Pathway for novel methodologies

2.2 FDA AI Credibility Framework (Jan 2025)

The FDA released a 7-step risk-based process for establishing AI model credibility:

1. **Define Context of Use (COU)** — what exactly will the model be used for?
2. **Assess Risk** — what’s the cost of model error?
3. **Establish Credibility Goals** — what level of evidence is needed?
4. **Choose Assessment Approaches** — independent validation, cross-validation, etc.
5. **Execute Assessment** — generate evidence
6. **Document** — transparent reporting
7. **Maintain** — monitor performance over time

Key insight: The FDA doesn’t require perfection — it requires that the model’s limitations are understood and documented relative to its context of use.

2.3 FDA/EMA “Guiding Principles of Good AI Practice” (Jan 2026)

10 principles emphasizing: - **Context of use** as the foundation for all validation - **Transparency** about data sources, model architecture, and limitations - **Reproducibility** — code and data should enable independent verification - **Bias assessment** across demographic sub-groups - **Maintenance plans** for models that may drift over time

2.4 What This Means

The regulatory path for digital twin-derived methodologies exists but is rigorous: - **PROCOVA** has the clearest path: it’s already accepted under both EMA and FDA - **Synthetic control arms** face higher scrutiny — they essentially replace randomization - **No agency has approved a true digital twin** for a primary endpoint analysis - The bar is lower for **ancillary/sensitivity analyses** and **trial design optimization**

Vendor Ecosystem

Figure 1: Vendor Ecosystem

3. Vendor Ecosystem

3.1 Market Structure

The digital twin space in clinical development has four segments:

3.2 Segment Profiles

Segment 1: DTaaS (Digital Twin as a Service)

Company	Core Offering	Technology	Regulatory Status	Key Limitations
Unlearn.AI (San Francisco)	Synthetic control arms via hierarchical Bayesian models predicting placebo outcomes	Neural Boltzmann Machines + PROCOVA; deep generative models on historical trial databases	EMA qualified (2022); FDA acknowledged (2024); PROCOVA covered under covariate adjustment guidance	Requires abundant historical placebo data → limited for novel mechanisms or pandemic scenarios. Statistical (not mechanistic) → less generalizable. PROCOVA assumes perfect calibration → simulations show failure under population shift. Not transparent about generative model architecture or validation data. Pricing: \$500K-\$2M per trial.
Altis Labs (Toronto)	AI-powered external control arms using imaging data matching	Imaging AI biomarker matching	Early-stage	Limited published validation. Presented at ISPOR 2025 for Phase 3 NSCLC. Still establishing evidence base.
Phesi (Boston/London)	Digital trial arm services, protocol optimization	Real-world clinical data matching to construct virtual patients	Proof-of-concept only (cGvHD digital twin published in Bone Marrow Transplantation, 2024)	Never used in a prospective regulatory submission. >2,000 virtual patients matched historical ORR (52.7%) but no regulatory precedent.

Segment 2: MIDD Software Platforms

Company	Offering	Revenue Model	Moat	Weakness
Certara	Pharmacometrics/QSP software (Phoenix, Simcyp)	Licenses + consulting + training	Deep regulatory relationships; EDC integration	Requires in-house pharmacometricians → poor accessibility for small biotechs
Simulations Plus	PBPK, PK/PD modeling (GastroPlus, DILIsym)	Licenses + consulting	Regulatory endorsement for PBPK	Same — requires specialized in-house expertise
Pumas-AI	Next-gen pharmacometrics (Julia-based, cloud-native)	Licenses + cloud	Modern architecture; tech-forward	Smaller ecosystem than incumbents
Rosa & Co	Quantitative systems pharmacology	Consulting + licenses	Mechanistic model expertise	Consulting-heavy model limits scalability

Segment 3: CRO-Embedded Modeling

Company	Offering	Revenue Model	Moat
IQVIA (MIDD practice)	Integrated modeling + clinical execution	\$500–\$2,000/hr for senior pharmacometricians, embedded in CRO contracts	Regulatory filing experience; proprietary historical databases
Parexel (Quantitative Solutions)	Pharmacometric consulting	Same	Same
Labcorp (Covance)	Integrated modeling services	Same	Same
PPD (Thermo Fisher)	Integrated modeling services	Same	Same

3.3 What the Vendors Are NOT Saying

Critical gaps that vendors underemphasize:

1. **No true digital twin exists for clinical trials.** Every “digital twin” is a static model trained on historical data. They are not dynamically updated with patient data during the trial.
2. **External validity is unknown.** Models trained on historical trial data may not generalize to populations with different demographics, standard of care, or enrollment criteria.
3. **Population shift is ignored.** PROCOVA (and most vendors) assume the prognostic model is well-calibrated for the trial population. Our simulations show this fails under even moderate shift.

Figure 2: Methodological Taxonomy

4. **Black-box models are the norm.** Most vendors don't disclose model architecture, training data provenance, or validation splits.
 5. **Cost savings are theoretical.** The \$500K-\$2M per-trial fee often exceeds the cost of simply enrolling more patients.
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4. Methodological Approaches

4.1 Taxonomy of Methods

4.2 PROCOVA in Detail

Since PROCOVA is the most prominently presented and marketed DT methodology, it deserves close examination.

How it works: 1. Historical data → train prognostic model → produces score $S(W)$ for any patient 2. Trial enrolls patients → baseline covariates collected 3. Each patient receives a prognostic score = predicted outcome under control 4. Primary analysis: Cox/ANCOVA with $S(W)$ as covariate

Strengths: - Statistically principled (special case of ANCOVA) - Preserves Type I error (by randomization, $S(W) \perp A$) - EMA qualified, FDA acknowledged - Flexible — any predictive model can generate the score

Limitations: - **Assumes perfect calibration.** If $S(W)$ is miscalibrated (population shift), power gains are reduced or reversed. - **No diagnostic.** No way to detect miscalibration from trial data. - **No adaptation.** The score is fixed — no mechanism to update it using trial data. - **Sensitive to strong confounders.** If unmeasured confounders differ, bias can be substantial.

4.3 MAP Prior Borrowing

The Meta-Analytic Predictive (MAP) prior approach (Neuenschwander et al., 2010) uses summary statistics from historical trials to construct an informative prior for the current trial's control group. The Robust MAP extension (Schmidli et al., 2014) adds a vague component via mixture to protect against prior-data conflict.

- Constructs a Bayesian prior from historical trial evidence (aggregate-level, not patient-level)
- Borrowing strength is controlled by the heterogeneity across historical trials — more consistent historical data yields stronger borrowing
- Does not require patient-level data from historical trials; operates on summary statistics
- Limited by commensurability — if historical and current trial populations differ, the vague component absorbs the conflict but power gains are reduced

4.4 Synthetic Control Arms (ECAs)

- Construct control cohort from external data using matching or weighting
- Regulatory acceptance is limited — FDA requires “compelling evidence” for replacing a control arm
- Major concerns: unmeasured confounding, temporal drift, selection bias

- Currently recommended for sensitivity/ancillary analyses, not primary inference

5. Evidence: What Has Actually Been Done?

5.1 Published Applications with Real Data

Study	Year	Method	Outcome	Strength of Evidence
Phesi cGvHD twin	2024	Digital twin from RWD	Matched historical ORR (52.7%)	Proof of concept, no prospective use
JHU Virtual Heart	2025	Mechanistic cardiac model	80% success (vs ~60% typical)	FDA pilot, n=10, promising but tiny
Altis MYSTIC ECA	2025	Imaging AI matching	ECA for Ph3 NSCLC	ISPOR poster, limited detail
Unlearn PROCOVA	2022-24	Bayesian + PROCOVA	Multiple trials, undisclosed	Limited public validation data
Internal calibration research	2026	Regularized score calibration	Simulation study	Ongoing internal work, manuscript in prep

Key observation: There is not a single published case of a digital twin methodology being used as the **primary analysis** in a successful regulatory submission. Every application is either a simulation, a proof of concept, or a sensitivity analysis.

5.2 Simulation Evidence

Two simulation studies are particularly relevant:

1. **Schuler et al. (2021)** — PROCOVA proof of concept. 1,000 reps, limited scenarios. Showed power gains under ideal conditions (no shift).
2. **Ongoing internal calibration research.** Preliminary simulation results suggest that regularized calibration of prognostic scores can recover most of the power lost to population shift, with minimal penalty when no shift is present. A manuscript is in preparation.

6. The Hype-Reality Gap

6.1 What the Hype Claims

Based on vendor marketing, industry press, and consulting reports:

“Digital twins will reduce trial timelines from 10-15 years to 5-7 years.”

“Synthetic control arms can replace 50% of placebo patients.”

“AI-powered digital twins predict patient outcomes with 95% accuracy.”

“Every patient can have a personalized digital twin.”

Hype vs Reality — Illustrative Assessment

Figure 3: Hype vs Reality — Illustrative Assessment

6.2 What the Evidence Actually Shows

Note on the figure below: The hype-reality comparison chart is a qualitative, illustrative assessment based on the author’s synthesis of the vendor claims and published evidence reviewed in this survey. The numerical values shown are not derived from a formal empirical study and should be interpreted as directional estimates for discussion purposes.

Claim	Reality Check
“Replace control arms”	No published regulatory submission has used a DT as primary analysis. FDA guidance requires “compelling evidence.”
“Reduce sample size by 50%”	PROCOVA simulations show ~15-30% reduction under ideal conditions. Under shift, gains can vanish.
“Predict individual outcomes”	Models predict average outcomes well but individual predictions have wide uncertainty intervals.
“Full physiological twin”	Roche’s Christodoulou (2025): “remains on the horizon.” Industry consensus: 5-7 years away.
“Works for any indication”	Most validation is in well-studied indications with abundant historical data (oncology, CNS).
“AI is the differentiator”	The statistical methodology matters more than the ML model. PROCOVA is essentially ANCOVA + a score.

6.3 Sources of Hype

1. **Vendor incentives.** Per-trial pricing (\$500K–\$2M) creates incentives to oversell capabilities.
2. **Regulatory ambiguity.** “FDA acknowledged” ≠ “FDA approved.” PROCOVA is “covered by existing guidance” — not a new qualification for synthetic control arms.
3. **Terminology inflation.** “Digital twin” sounds more impressive than “prognostic score adjustment” or “external control arm.”
4. **Consulting industry.** McKinsey, BCG, Deloitte all have pharma AI practices that amplify the narrative.

6.4 A Realistic Assessment

What digital twins CAN do today:

- Improve power in well-understood indications with abundant historical data
- Supplement (not replace) randomized control arms
- Enable adaptive trial designs by simulating enrollment scenarios
- Identify enrichment strategies and patient subgroups

What digital twins CANNOT do today:

- Replace a randomized control arm as the sole basis for primary inference
- Predict individual patient outcomes with clinically actionable precision
- Generalize safely to populations with different standard of care
- Work for novel mechanisms with no historical precedent

7. Practical Considerations for Internal Use

7.1 Calibration of Vendor Scores Using Internal Data

A recurring theme across the digital twin landscape is the assumption that vendor-provided prognostic scores are well-calibrated for the target trial population. In practice, this assumption is often violated. Population shift between the historical data used to build a vendor's model and the actual trial — due to evolving standard of care, changing eligibility criteria, biomarker drift, or simple demographic differences — can substantially reduce the efficiency gain that PROCOVA or similar methods promise.

A pragmatic approach is to treat any externally-provided prognostic score as a starting point rather than a final product. The score can be evaluated and, if needed, fine-tuned using internal data:

- **Within a trial (blinded).** A calibration model can be fitted on the trial's own blinded data to detect and correct miscalibration. This approach preserves trial integrity (no unblinding needed) and tailors the score to the actual enrolled population. The ridge penalty protects against overfitting when the calibration sample is limited.
- **Outside a trial (historical internal data).** If a company has a rich repository of completed trial data, it can pre-calibrate vendor scores against its own internal benchmark before deploying them in new trials.

Neither approach requires trusting the vendor's claims about calibration. Both leverage the one asset that large pharmaceutical companies have that vendors typically do not: extensive, internally-validated clinical trial data.

7.2 Ongoing Internal Work

Work is underway to develop and validate this calibration framework.

8. Strategic Implications

8.1 Key Takeaways

1. **The “digital twin” label is heavily marketed** but the actual methods are well-understood statistical techniques (prognostic score adjustment, synthetic control arms, Bayesian borrowing). Don't let terminology drive strategy.
2. **Vendor scores should be treated as inputs, not oracles.** Any score — from Unlearn, Altis, Phesi, or an academic group — can be miscalibrated for a specific trial population. Internal validation and calibration are essential.
3. **Your own data is your biggest asset.** Merck's repository of completed trial data is something no vendor can match. Using it to validate, calibrate, or challenge vendor claims is the most defensible strategy.
4. **Blinded calibration within a trial is the cleanest approach.** It avoids unblinding, preserves trial integrity, and addresses population shift at the point of use.
5. **Don't overpay for black-box services.** Full-service “digital twin” offerings at \$500K–\$2M per trial are hard to justify when simpler statistical methods achieve comparable gains.

8.2 Potential Next Steps

Action	Timeline	Effort	Notes
Internal calibration pilot	2-3 months	Medium	Retrospective on completed trials
Vendor score evaluation framework	1 month	Low	Standardized validation protocol
Engagement with regulators	6-12 months	Medium	MIDD meeting for calibration methodology
Internal capability building	3-6 months	Medium	R package, SOP, training

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End of landscape survey. Draft for discussion.